Volatility Trading with Delta-Hedged ATM Straddles

Dingjingmu (Steven) Yang
STAT8020: Quantitative Strategies and Algorithmic Trading
Master of Data Science
The University of Hong Kong
u3664645@connect.hku.hk

Abstract

This report investigates a systematic volatility-trading strategy that uses delta-hedged, one-month at-the-money (ATM) straddles across a diverse universe of U.S. ETFs and individual equities. By evaluating four signaling mechanisms—ranging from fixed and percentile-based volatility risk premium (VRP) thresholds to GJR-GARCH(1,1) conditional forecasts—the study identifies the long-horizon realized-volatility rule as the most robust signal for capturing the volatility risk premium across asset classes. Empirical results show that these signals effectively isolate volatility mispricing, with PnL attribution confirming that returns are consistently driven by vega exposure rather than by underlying price direction. This stability across both broad-market indices and single stocks suggests that the framework provides a reliable basis for identifying entry points where implied volatility deviates significantly from realized expectations.

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1 Introduction

This report examines a low-frequency volatility-trading strategy implemented on U.S. equities and exchange-traded funds. Volatility trading matters because uncertainty is itself priced in financial markets. Investors care not only about expected returns, but also about the dispersion and timing of future price movements, especially when portfolios are exposed to crash risk, hedging demand, or large mark-to-market losses. Options therefore allow future uncertainty to be traded more directly than the underlying asset alone.

At a broad level, volatility trading is profitable when the price paid for future uncertainty differs from the uncertainty that is eventually realized. If option-implied volatility is too high relative to subsequent realized volatility, selling volatility can be profitable because the market has overcharged for protection or convexity. If option-implied volatility is too low relative to subsequent realized volatility, buying volatility can be profitable because the option market has underpriced future movement. More generally, profit opportunities can arise when implied volatility differs from a suitable benchmark for expected volatility, when investors are willing to pay a premium for downside insurance, or when compensation for tail risk and skewness is embedded in option prices [1, 2].

2 Options Preliminaries

2.1 Call, put, and straddle

A call option gives its holder the right, but not the obligation, to buy the underlying asset at a predetermined strike price K before or at expiry. A put option gives the right to sell at the strike price K. The holder of a long option pays the option price upfront and has limited downside equal to that payment. The seller of the option receives that payment upfront but assumes the contingent liability.

At expiry T, the terminal payoffs of a call and a put are

$$\text{Call payoff} = \max(S_T - K, 0), \qquad \text{Put payoff} = \max(K - S_T, 0),$$

where S_T is the underlying price at expiry.

The strategy trades an at-the-money straddle, which is the combination of one call and one put written simultaneously on the same underlying, with the same strike and the same expiry. At expiry, the payoff of a long straddle is

$$\max(S_T - K, 0) + \max(K - S_T, 0) = |S_T - K|,$$

so the position benefits from sufficiently large moves in either direction. This makes the straddle the natural vehicle for expressing a view on volatility rather than on market direction.

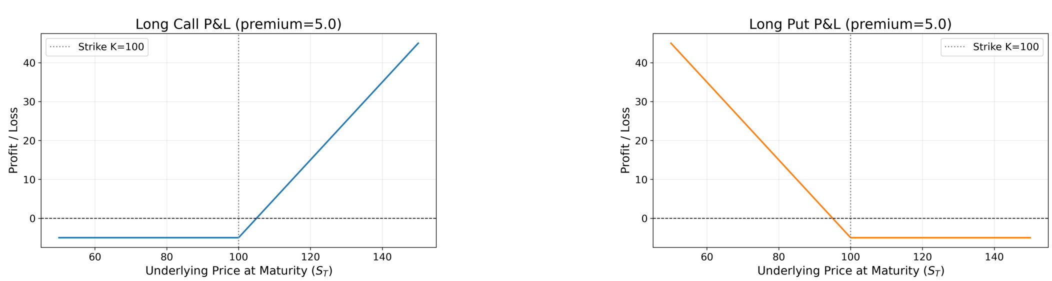

Figure 1: Illustrative maturity PnL of a long call and a long put.

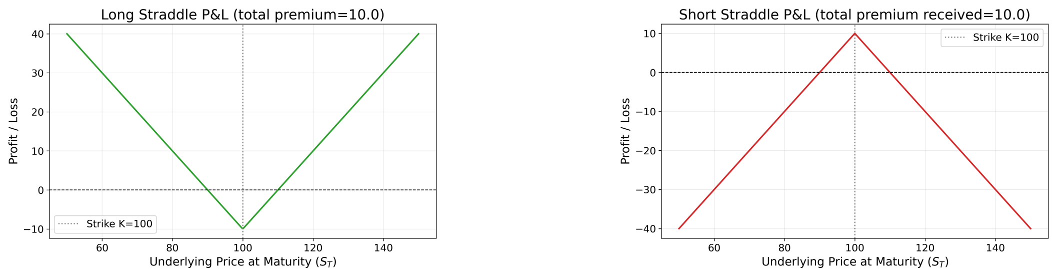

Figure 2: Illustrative maturity PnL of long and short ATM straddles.

2.2 Black–Scholes model

The Black–Scholes model is used as the starting point for option valuation. It provides a tractable mapping between an option's value and the key state variables that drive it: the underlying price, strike, time to maturity, risk-free rate, and volatility. Although listed equity options are American-style in legal form, Black–Scholes remains a standard and useful approximation for short-dated liquid contracts, especially when the main objective is to infer implied volatility and measure risk exposures consistently.

For a European call and put with time to maturity $\tau = T - t$, risk-free rate r, and volatility σ, prices are

$$C = S_t N(d_1) - K e^{-r\tau} N(d_2), \qquad P = K e^{-r\tau} N(-d_2) - S_t N(-d_1),$$

where

$$d_1 = \frac{\ln(S_t/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau},$$

and $N(\cdot)$ denotes the standard normal cumulative distribution function.

2.3 Greeks

The value of an option depends on the underlying price, implied volatility, time to expiry, and interest rates. The Greeks summarize local sensitivities:

$$\Delta = \frac{\partial V}{\partial S}, \qquad \Gamma = \frac{\partial^2 V}{\partial S^2}, \qquad \Theta = \frac{\partial V}{\partial t}, \qquad \nu = \frac{\partial V}{\partial \sigma}, \qquad \rho = \frac{\partial V}{\partial r},$$

$$\text{Vanna} = \frac{\partial^2 V}{\partial S\, \partial \sigma}, \qquad \text{Volga} = \frac{\partial^2 V}{\partial \sigma^2}.$$

For the present strategy, the most important interpretations are straightforward. Delta measures first-order directional exposure to the underlying. Gamma captures convexity in the underlying price. Vega measures sensitivity to changes in implied volatility. Theta represents time decay. Vanna and volga become relevant when the underlying and implied volatility move jointly or when volatility changes materially.

A second-order expansion of option value over a short interval provides the basic PnL map:

$$\mathrm{d}V \approx \Delta\,\mathrm{d}S + \frac{1}{2}\Gamma\,\mathrm{d}S^2 + \nu\,\mathrm{d}\sigma + \Theta\,\mathrm{d}t + \rho\,\mathrm{d}r + \text{Vanna}\,\mathrm{d}S\,\mathrm{d}\sigma + \frac{1}{2}\text{Volga}\,(\mathrm{d}\sigma)^2 + \varepsilon,$$

where ε captures higher-order terms. This decomposition follows the standard treatment of option risk attribution in [3]. It is useful later when interpreting whether a delta-hedged straddle is actually earning volatility-related PnL.

2.4 Implied volatility

Implied volatility is the value of σ that makes the Black–Scholes price match the observed market price of an option or option package. In that sense, implied volatility is not directly observed; it is inferred from price. Practitioners often quote options in implied-volatility terms because volatility provides a standardized way to compare option richness across time and across contracts.

This interpretation is central to the strategy studied in this report. The trading decision is not driven by a view on whether the stock will rise or fall. Instead, it is driven by whether the option market appears to be charging too much or too little for future uncertainty relative to a benchmark for expected or normal volatility.

3 Methodology

3.1 Capturing volatility mispricing

The central implementation question is how to convert a view on volatility into a tradable position. The basic idea is to compare the option market's price of future uncertainty, summarized by implied volatility, with a benchmark for the volatility expected to materialize. When implied volatility is meaningfully above that benchmark, volatility can be viewed as expensive and the natural trade is to sell volatility. When implied volatility is meaningfully below that benchmark, volatility can be viewed as cheap and the natural trade is to buy volatility. In this sense, the strategy seeks to capture volatility mispricing by trading the gap between implied volatility and realized- or forecast-volatility measures.

A straddle is a natural instrument for this purpose because its payoff depends primarily on the magnitude of price movement rather than its direction. A long straddle benefits from high volatility because sufficiently large realized moves increase the value of the embedded convexity. By contrast, a short straddle benefits from low volatility because the option seller receives premium upfront and profits when realized price movement remains limited relative to what the option market had priced in. This also explains why the absolute level of implied volatility matters. Even if an investor expects volatility to be elevated, a long straddle may still be unattractive when implied volatility is already very high, because the option premium may already embed that expectation and leave too little room for additional profit.

However, simply buying or selling a straddle does not isolate a pure volatility exposure. The position still carries delta, so part of its profit and loss may come from directional movement in the underlying. To make the trade more interpretable as a volatility position, the strategy uses delta hedging. Operationally, this means trading the underlying stock against the straddle so that net portfolio delta remains close to zero. The goal is to reduce first-order directional exposure from delta and allow realized profit and loss to depend more cleanly on convexity, time decay, and changes in implied volatility.

In practice, repeated delta hedging is closely related to gamma scalping. A long-gamma position that is rebalanced toward delta neutrality tends to buy the underlying after price declines and sell after price increases. If realized price fluctuations are sufficiently large, this rebalancing process can monetize convexity gains and offset part of the theta paid to hold the options [4]. Conversely, if realized volatility remains too low, theta decay dominates and the long-volatility position loses money. This gamma-scalping interpretation matters because it clarifies that delta hedging is not merely a risk-control overlay; it is also part of the mechanism through which a volatility position converts realized market movement into profit and loss.

For a long-gamma trade, one therefore wants the *gamma* contribution to dominate the *theta* drag over a typical rebalancing interval. Holding other risks fixed, the local second-order expansion suggests incremental P&L of the form $\frac{1}{2}\Gamma\,(\mathrm{d}S)^2 + \Theta\,\mathrm{d}t$ for the option leg (with $\Gamma > 0$ for a long straddle and $\Theta < 0$ under the usual sign convention). The gamma–theta breakeven in *spot* space is obtained by equating the magnitude of convexity P&L to time decay, $|\frac{1}{2}\Gamma\,(\mathrm{d}S)^2| \approx |\Theta|\,\mathrm{d}t$, which yields a characteristic move scale $|\mathrm{d}S| \gtrsim \sqrt{2\,|\Theta|\,\mathrm{d}t/\Gamma}$. Because $\mathrm{d}\Delta \approx \Gamma\,\mathrm{d}S$ to leading order, the same breakeven logic can be expressed in *delta* space as $|\mathrm{d}\Delta| \gtrsim \sqrt{2\,|\Theta|\,\Gamma\,\mathrm{d}t}$. In the data, Θ is quoted per one *year* of time-to-expiry (the same 252-trading-day convention used for P&L time steps), so for a one-trading-day rehedge check we set $\mathrm{d}t = 1/252$. The implementation uses a band $k\sqrt{2\,|\Theta|\,\Gamma/252}$ on the absolute change in net delta since the last hedge, with Θ and Γ taken from the current bar of the traded straddle, and position-specific multipliers: $k \geq 1$ when long gamma (allow a wider no-trade region so realized convexity can work against theta before rebalancing) and $k < 1$ when short gamma (a comparatively tighter band, reflecting that adverse gamma should be cut sooner relative to the same breakeven scale).

3.2 Trading signal

The core mapping from signal to trade direction is simple. When implied volatility is high relative to the benchmark, the straddle is treated as expensive, so the strategy takes a short-volatility position by selling the straddle, and vice versa. Positions are closed when the signal reverts toward its neutral region, and the hedge is updated whenever the portfolio's net delta drifts beyond the rebalancing threshold.

3.2.1 Realized volatility benchmark

The classic benchmark in volatility trading is the volatility risk premium (VRP) [1],

$$\mathrm{VRP}_t = \mathrm{IV}_t - \mathrm{RV}_t.$$

Here, IV_t denotes the implied volatility of the traded straddle and RV_t denotes realized volatility estimated from recent underlying returns. This spread is widely used to capture the variance risk premium embedded in option prices. At the same time, the measured spread need not reflect pure volatility mispricing alone: part of it may also compensate investors for higher-order risks such as downside asymmetry and jump-related skewness, especially for individual equities with more pronounced tail risk [2]. The intuition is direct: if implied volatility is high relative to realized volatility, the option market may be overpricing future uncertainty; if implied volatility is low relative to realized volatility, options may be relatively cheap.

Realized volatility is computed from a rolling 20-day window of daily log returns. If $r_t = \ln(S_t/S_{t-1})$, then

$$\mathrm{RV}_{t,20} = \sqrt{252}\sqrt{\frac{1}{20}\sum_{i=1}^{20}(r_{t+1-i} - \bar{r}_t)^2},$$

where $\bar{r}_t$ is the mean return over the same window.

This specification is the most familiar and transparent benchmark in the report because it compares option-implied uncertainty with the variability recently observed in the underlying market.

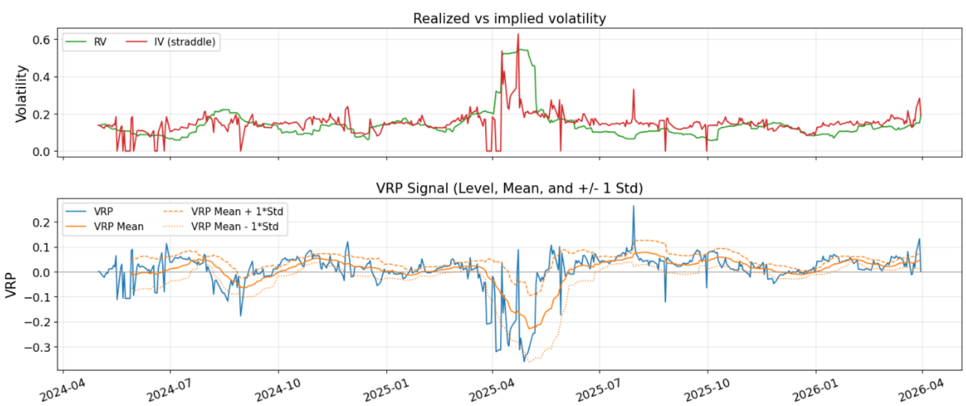

Figure 3: SPY implied volatility and realized volatility in the upper panel, and the corresponding volatility risk premium in the lower panel. The lower panel also reports the rolling mean together with mean ± 1 standard deviation bands.

3.2.2 GARCH-based benchmark

The second benchmark addresses an important mismatch in a plain VRP comparison. Implied volatility reflects the market's expectation of volatility from the current date to option expiry, whereas realized volatility is computed from historical returns and need not persist into the future. In other words, the standard VRP compares a backward-looking statistic with a forward-looking market expectation.

To make the comparison more consistent in horizon, this benchmark replaces backward-looking realized volatility with a model-based forecast from a GJR-GARCH(1,1) specification with Student-t

innovations. This choice allows for volatility clustering and asymmetric responses to negative returns, both of which are common in equity data [5]. The model is estimated on a trailing window, refit periodically, and projected forward over the remaining trading days to the option expiry.

The corresponding spread is

$$\mathrm{IV}_t - \widehat{\mathrm{RV}}_t^{\mathrm{GARCH}}.$$

This benchmark compares current option-implied volatility with a forward-looking conditional volatility estimate rather than with a purely historical measure. Operationally, the signal asks whether market-implied future volatility is aligned with model-predicted future volatility.

3.2.3 Long-horizon realized-volatility benchmark

A third benchmark compares current straddle implied volatility with a slow-moving historical average of realized volatility. Specifically, it uses the mean of the prior 60 observations of daily realized volatility as a long-horizon anchor. This choice is motivated by the well-known persistence of volatility together with its tendency to revert toward more normal levels over time, as discussed by Engle and Patton [6]. The corresponding spread is therefore

$$\mathrm{IV}_t - \overline{\mathrm{RV}}_t^{\mathrm{long}},$$

where $\overline{\mathrm{RV}}_t^{\mathrm{long}}$ is computed from past values only.

The economic logic follows a sequence of convergence arguments grounded in both theory and empirical regularities. As expiry approaches, option-implied volatility should converge toward realized volatility. A second empirical regularity is that realized volatility itself often mean-reverts [6, 7]. Combining these ideas yields a practical hierarchy: implied volatility converges toward realized volatility, and realized volatility converges toward its longer-run mean. Under this view, implied volatility should also tend to gravitate toward long-horizon realized volatility as expiry approaches.

3.2.4 Z-score normalization

The trading logic compares current straddle implied volatility with these benchmarks through a rolling 20-day z-score,

$$Z_t(X) = \frac{X_t - \mu_{t,20}(X)}{\sigma_{t,20}(X)},$$

where X_t denotes the spread of interest and $\mu_{t,20}(X)$ and $\sigma_{t,20}(X)$ are its rolling mean and standard deviation. A sufficiently negative z-score indicates that options appear cheap relative to the benchmark, while a sufficiently positive z-score indicates that options appear expensive. This normalization makes the trading threshold comparable across different volatility regimes.

3.2.5 Signal summary

The implementation includes four signal rules built on these benchmarks. The hard-threshold rule and the percentile rule both use the classic VRP idea, but they differ in how entry thresholds are defined. The GARCH-based rule replaces realized volatility with a conditional forecast, whereas the long-horizon rule compares current implied volatility with a slow-moving historical mean of realized volatility.

Table 1: Summary of trading rules.

Agent	Long-volatility entry	Short-volatility entry	Exit rule
Hard-threshold VRP	$Z_t(\mathrm{IV} - \mathrm{RV}) < -k$	$Z_t(\mathrm{IV} - \mathrm{RV}) > k$	z-score crosses zero
Percentile VRP	rolling $Z_t(\mathrm{IV} - \mathrm{RV})$ in lower empirical tail	rolling $Z_t(\mathrm{IV} - \mathrm{RV})$ in upper empirical tail	z-score crosses historical median
GARCH spread	$Z_t(\mathrm{IV} - \widehat{\mathrm{RV}}^{\mathrm{GARCH}}) < -c$	$Z_t(\mathrm{IV} - \widehat{\mathrm{RV}}^{\mathrm{GARCH}}) > c$	z-score crosses zero
Long-horizon RV spread	$Z_t(\mathrm{IV} - \overline{\mathrm{RV}}^{\mathrm{long}}) < -c$	$Z_t(\mathrm{IV} - \overline{\mathrm{RV}}^{\mathrm{long}}) > c$	z-score crosses zero

The reported implementation uses the parameter settings summarized below.

Table 2: Shared execution parameters.

Parameter	Description	Value
`allow_short`	Whether short-volatility positions are allowed	True
`delta_hedge`	Whether the straddle is hedged with the underlying	True
`long_rehedge_threshold`	Rehedge band multiplier k for long straddle ($k\sqrt{2\lvert\Theta\rvert\Gamma/252}$, Θ per year of T; current bar)	1.2
`short_rehedge_threshold`	Same multiplier k when short straddle	0.8

Table 3: Agent-specific signal parameters.

Agent	Parameter	Value
Hard-threshold VRP	k	1.0
Percentile VRP	Entry percentile	0.20
GARCH spread	Entry threshold c	1.0
Long-horizon IV spread	Entry threshold c	1.0
Long-horizon IV spread	Long-term RV window	60

All parameters are fixed before the backtest is run. They are not fitted, trained, or optimized in sample, and the same parameter set is applied to every underlying in the sample. This design limits overfitting and makes the cross-asset comparison more meaningful.

4 Data

The empirical sample consists of three exchange-traded funds—SPY, QQQ, and DIA—together with a set of individual U.S. equities: AAPL, ACMR, AMD, GOOG, INTC, META, and TSLA; the backtest is conducted from 2024 May 1st to 2026 March 31st. These assets provide a useful mix of broad market exposure and single-name volatility, allowing the strategy to be examined across instruments with different liquidity conditions, sector compositions, and event risk. The individual stocks are not selected through a formal screening rule; rather, they form an illustrative sample of liquid and well-known names rather than a comprehensive cross-section of U.S. equities.

Although the strategy is implemented on all of these symbols, the report focuses primarily on SPY. The reason is both practical and economic: SPY is the most liquid ETF in the sample; its options market is especially deep, and its prices are therefore less affected by idiosyncratic microstructure noise than those of many single stocks. SPY thus provides the cleanest setting in which to evaluate whether the strategy behaves as intended.

For each symbol and each month, the strategy selects an at-the-money straddle associated with the next monthly expiry. The strike is fixed at entry and the position is monitored through the month. As the underlying moves, the contract drifts away from exact at-the-money status; this is a deliberate simplification consistent with a low-frequency monthly trading design.

4.1 Data construction details

Each daily observation combines the underlying close, cash dividends, a short-dated risk-free rate, and the closing prices of the selected call and put. From these inputs, the dataset computes realized volatility, leg-level implied volatilities, a straddle implied volatility that matches the observed straddle price, and the standard Greeks for both the individual legs and the straddle as a whole. All volatility measures are annualized using a 252-trading-day convention.

5 Results

5.1 Performance on SPY

5.1.1 Return summary

Table 4: SPY performance summary under updated parameter settings.

Agent	Win Rate	Sharpe	Sortino	Ann. Return	Ann. Vol.	Max DD	Calmar	Kelly
`Agent_hardThreshold`	75.00%	1.3248	1.1919	25.96%	17.42%	-11.12%	2.3339	7.6042
`Agent_Perc`	74.36%	1.2669	1.1843	24.85%	17.52%	-11.12%	2.2345	7.2312
`Agent_Garch`	74.47%	1.0765	0.9319	21.49%	18.09%	-14.99%	1.4338	5.9520
`Agent_LongTerm`	73.91%	1.0618	1.1015	25.56%	21.44%	-11.95%	2.1381	4.9534

Table 4 summarizes the frictionless backtest and shows strong performance across all four agents. However, these results assume zero execution costs and should therefore be interpreted as an upper bound on implementable performance.

`Agent_hardThreshold` delivers the strongest risk-adjusted results, with the highest Sharpe 1.32, Sortino 1.19, and Calmar 2.33 ratios, and the smallest maximum drawdown -11.12%, making it the most consistent performer across all metrics. `Agent_Perc` is closely comparable, posting the same drawdown and very similar volatility figures with only marginally lower returns, suggesting that the percentile-based signal and the fixed-threshold signal capture a very similar set of trading opportunities. `Agent_Garch` produces the lowest annualized return 21.49% and the largest drawdown -14.99%, likely because the GARCH forecast introduces additional estimation lag that causes the signal to react more slowly to sharp volatility regimes. `Agent_LongTerm` achieves a competitive annualized return 25.56% but does so with notably higher volatility 21.44%, reflecting the wider swings inherent in a long-horizon realized-volatility benchmark that is slower to adapt to short-term market conditions.

Figure 4 reports cumulative value-ratio paths (top panel) and daily return series (bottom panel) for the frictionless SPY backtest. These paths appear attractive in isolation, but the comparison with Table 7 shows that the apparent edge is not robust once realistic execution frictions are introduced.

Figure 4: Cumulative value ratio (top) and daily returns (bottom) for each agent on SPY. Paths are flat initially while each strategy completes its signal warm-up: most agents need 20 completed trading days before thresholds are available, and the long-horizon agent needs 60.

5.1.2 PnL attribution

Daily PnL is decomposed via a second-order Greek expansion:

$$\mathrm{d}V \approx \underbrace{\Delta\,\mathrm{d}S}_{\text{Delta}} + \underbrace{\frac{1}{2}\Gamma\,\mathrm{d}S^2}_{\text{Gamma}} + \underbrace{\nu\,\mathrm{d}\sigma}_{\text{Vega}} + \underbrace{\Theta\,\mathrm{d}t}_{\text{Theta}} + \underbrace{\rho\,\mathrm{d}r}_{\text{Rho}} + \underbrace{\text{Vanna}\,\mathrm{d}S\,\mathrm{d}\sigma}_{\text{Vanna}} + \underbrace{\frac{1}{2}\text{Volga}\,(\mathrm{d}\sigma)^2}_{\text{Volga}} + \varepsilon.$$

PnL from delta re-hedging trades is merged into the Gamma term, so the Gamma attribution captures both the convexity gain and the realized cost or benefit of maintaining the hedge.

For each Greek component i, the reported percentage contribution is computed as

$$\text{Greek }\%_i = \frac{\text{Greek}_i \text{ summed attribution}}{\sum_j |\text{Greek}_j \text{ summed attribution}|} \times 100.$$

Figure 5 shows the summed attribution breakdown for each agent. Vega is the dominant PnL source across all strategies, confirming that returns are driven primarily by volatility exposure rather than by directional market moves. Notably, `Agent_LongTerm` records the highest vega attribution (58.2%), indicating that the long-horizon realized-volatility signal is particularly effective at capturing volatility mispricing and subsequent implied-volatility mean reversion.

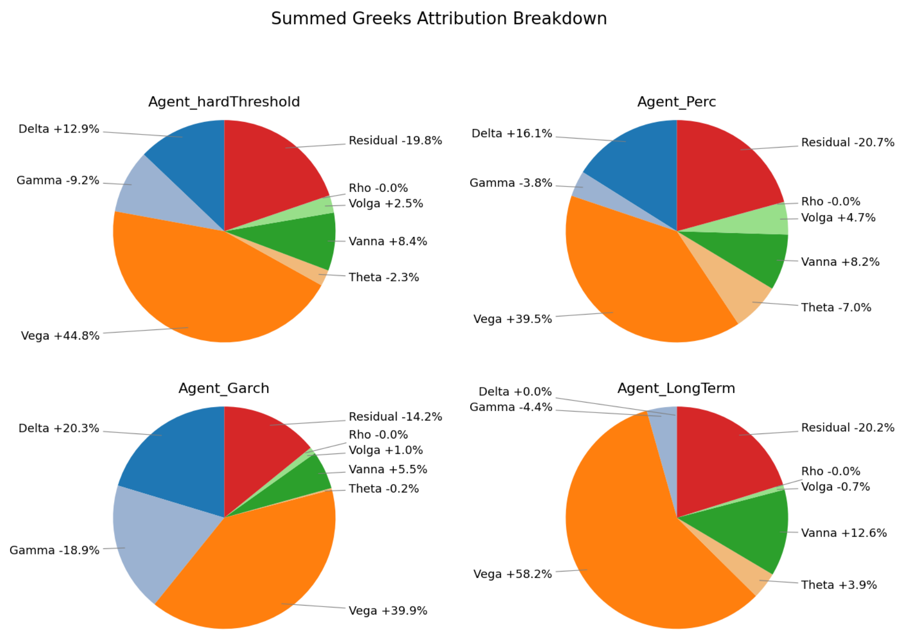

Figure 5: Summed Greeks attribution breakdown by strategy agent on SPY.

5.2 Performance on other ETFs

Table 5 reports performance statistics for QQQ and DIA under the same parameter set used in the SPY analysis. DIA is generally stronger on a risk-adjusted basis across agents, with higher Sharpe and Sortino ratios, whereas QQQ remains profitable but exhibits larger drawdowns and greater dispersion across specifications.

Table 5: QQQ and DIA performance summary by agent.

Ticker	Agent	Win Rate	Sharpe	Sortino	Ann. Return	Ann. Vol.	Max DD
QQQ	`Agent_hardThreshold`	81.08%	0.8870	0.6754	21.10%	21.58%	-20.95%
QQQ	`Agent_Perc`	69.23%	0.9749	1.2191	34.63%	30.50%	-22.95%
QQQ	`Agent_Garch`	76.32%	1.1106	0.9158	28.83%	22.81%	-19.81%
QQQ	`Agent_LongTerm`	80.00%	1.7709	1.5282	46.30%	21.48%	-14.83%
DIA	`Agent_hardThreshold`	87.50%	1.5611	1.2112	47.21%	24.77%	-20.59%
DIA	`Agent_Perc`	87.04%	2.2303	2.2453	91.18%	29.06%	-18.77%
DIA	`Agent_Garch`	85.45%	1.9131	1.7752	68.01%	27.12%	-18.97%
DIA	`Agent_LongTerm`	85.71%	2.0476	1.7590	81.61%	29.14%	-19.89%

Table 6 reports summed Greek attribution percentages. Across QQQ and DIA, Vega is the largest positive contributor for every agent, consistent with a volatility-risk-premium strategy. The strongest Vega attribution appears in `Agent_LongTerm` for both ETFs (QQQ: +57.72%, DIA: +66.42%), indicating that this signal captures volatility-premium exposure most directly in the ETF sample. Residual attribution is generally negative, which suggests that a non-trivial share of realized PnL lies outside the low-order Greek decomposition.

Table 6: QQQ and DIA Greeks attribution percentages by agent.

Ticker	Agent	Delta	Gamma	Vega	Theta	Vanna	Volga	Rho	Residual
QQQ	`Agent_hardThreshold`	+2.81%	-12.76%	+54.67%	-0.76%	+2.73%	+9.64%	+0.00%	-16.62%
QQQ	`Agent_Perc`	+10.38%	+9.51%	+19.51%	-12.03%	+6.83%	+15.87%	-0.00%	-25.88%
QQQ	`Agent_Garch`	-5.33%	+8.92%	+47.41%	-6.11%	+2.40%	+9.83%	+0.01%	-19.98%
QQQ	`Agent_LongTerm`	-0.77%	-19.46%	+57.72%	+3.42%	+2.34%	+7.13%	+0.02%	-9.14%
DIA	`Agent_hardThreshold`	-4.10%	+3.97%	+48.70%	-10.06%	+6.17%	+14.41%	+0.00%	-12.58%
DIA	`Agent_Perc`	-3.75%	+2.60%	+50.89%	-8.11%	+4.56%	+16.20%	-0.00%	-13.88%
DIA	`Agent_Garch`	-8.63%	+6.18%	+58.07%	-8.56%	+2.44%	+9.69%	+0.00%	-6.43%
DIA	`Agent_LongTerm`	-6.71%	+1.86%	+66.42%	+0.11%	+1.02%	+7.41%	+0.01%	-16.46%

5.3 Performance on single stocks

Single-stock results are more heterogeneous than ETF results, reflecting larger idiosyncratic risk, event sensitivity, and firm-specific jumps. In risk-adjusted terms, outcomes differ substantially across tickers: GOOG and INTC are comparatively strong across multiple specifications, whereas AMD and parts of the ACMR and META results are weaker or less stable in this sample.

From a Greeks-attribution perspective, vega remains the dominant positive component in most ticker-agent combinations. The highest Vega attribution in the single-stock panel is `Agent_Garch` on ACMR (+67.53%), while `Agent_LongTerm` also shows a high Vega share on INTC (+42.03%) and TSLA (+41.71%). This supports the same interpretation as in the ETF results: realized strategy PnL is primarily tied to volatility exposure, with residual and cross-Greek terms explaining the remainder.

The full single-stock statistics and Greeks-attribution tables are reported in Appendix A.

5.4 Robustness

The cross-asset results indicate that the long-horizon realized-volatility spread is the most robust signal in the study. Using a single fixed parameter set, without ticker-specific tuning, it delivers the most consistent risk-adjusted performance across SPY, QQQ, and DIA, while remaining mixed but still competitive in several single stocks.

The Greeks-based attribution supports the same conclusion: vega is the dominant positive contributor in most cases, indicating that strategy returns are linked primarily to volatility exposure rather than to incidental directional beta.

Robustness is stronger in ETFs than in single names, likely because ETFs benefit from better liquidity and lower idiosyncratic jump risk. The main limitations remain sample length, execution-cost assumptions, and residual components that lie outside the low-order Greek decomposition.

6 Practical Challenges and Limitations

6.1 Execution Frictions

Slippage is a major practical challenge because executed prices may differ from quoted prices, directly reducing realized profits. The frictionless backtests should therefore be interpreted as signal-quality benchmarks rather than as directly tradable results. To assess this effect, the report also tests a 0.1% slippage assumption.

Table 7: SPY performance summary after incorporating slippage.

Agent	Win Rate	Sharpe Ratio	Sortino Ratio	Annual Return	Annual Volatility	Max Drawdown	Calmar Ratio	Kelly's Criteria
`Agent_hardThreshold`	65.00%	-0.9852	-1.0949	-14.12%	15.45%	-26.89%	-0.5251	-6.3770
`Agent_Perc`	64.10%	-0.9934	-1.1351	-14.36%	15.60%	-27.47%	-0.5226	-6.3667
`Agent_Garch`	65.96%	-1.0275	-1.0072	-15.89%	16.84%	-33.34%	-0.4765	-6.1023
`Agent_LongTerm`	65.22%	-0.5136	-0.6959	-9.05%	18.46%	-21.91%	-0.4128	-2.7823

This strategy requires frequent delta hedging and therefore repeated buying and selling of the underlying asset. Even small execution shortfalls accumulate quickly when hedge rebalancing is frequent. This logic is consistent with the broader evidence in [8], which shows that illiquidity and trading costs are economically important determinants of realized returns. Under the 0.1% slippage assumption, the SPY results in Table 7 deteriorate sharply: all four agents move from attractive frictionless performance to negative annual returns, negative Sharpe ratios, and materially worse drawdowns. The practical implication is clear: unless execution costs can be reduced substantially, the strategy is not economically viable. A visual summary is provided in Appendix B.

6.2 Implementation Challenges

The trading signals themselves are straightforward to compute, but implementing the strategy in live markets is considerably more demanding.

As discussed above, maintaining a frequently rebalanced delta hedge is costly and operationally demanding, especially for individual investors. To illustrate the trade-off, the report also evaluates a pure unhedged straddle strategy on SPY.

Table 8: SPY performance summary for the unhedged straddle strategy.

Agent	Win Rate	Sharpe Ratio	Sortino Ratio	Annual Return	Annual Volatility	Max Drawdown	Calmar Ratio	Kelly's Criteria
`Agent_hardThreshold`	70.00%	1.2646	1.3248	579.74%	151.55%	-72.31%	8.0176	0.8345
`Agent_Perc`	69.23%	1.0237	1.1120	390.01%	155.24%	-72.31%	5.3938	0.6594
`Agent_Garch`	68.09%	1.6241	1.6751	1127.63%	154.40%	-70.02%	16.1035	1.0519
`Agent_LongTerm`	69.57%	0.5235	0.4641	130.97%	159.91%	-91.09%	1.4378	0.3274

Table 8 shows that removing the hedge can produce very large returns in this backtest, especially for `Agent_Garch`. However, these returns come at the cost of a dramatic increase in risk: annual volatility rises to roughly 150%–160%, and maximum drawdowns remain extremely severe, reaching -91.09% for `Agent_LongTerm`. In practice, the unhedged version is difficult to implement because the strategy is no longer close to a pure volatility trade; it carries substantial directional exposure and becomes much harder to size, finance, and hold through adverse market moves. A visual summary is provided in Appendix C.

Prior evidence also suggests that index straddles are often expensive on average, so selling them can be profitable in expectation. Coval and Shumway [9] show that short S&P 500 straddle positions earn positive average returns, which is consistent with the volatility risk premium idea used in this report. However, that profitability comes with extreme tail risk: a short straddle is highly exposed to large underlying moves, so losses can become very large, especially when the position is left unhedged. In practice, short-straddle positions also require substantial margin. When that capital is locked up as collateral, the strategy imposes a meaningful opportunity cost, because the same funds cannot be deployed elsewhere during the holding period. An alternative way to express a short-volatility view is to use a butterfly option structure to manage tail risk; although this approach is not discussed in detail in this report, it is a practical way to short volatility with more limited downside exposure.

6.3 Limitations

This backtesting framework has several important limitations. First, the options traded in practice are not always exactly at the money. Because implied volatility is typically skewed across strikes, the implied volatility used to construct the trading signal may not be fully comparable across contracts. Part of the measured signal may therefore reflect strike-specific volatility skew rather than a clean mispricing of aggregate volatility.

Second, the delta-hedging overlay introduces an implicit gamma-scalping component. Although the hedge is adjusted for risk management rather than for directional trading, rebalancing mechanically tends to buy the underlying after price declines and sell after price increases. This effect is strongest when the option remains close to the money, where gamma is largest. As the traded contract drifts away from the strike, gamma declines, so the realized benefit from this passive gamma-scalping mechanism is weaker than an idealized at-the-money strategy might suggest.

A further limitation is theta decay. As remaining time to maturity becomes short, especially below roughly two weeks, time decay can become substantial and quickly erode the value of long-volatility positions.

In principle, many of these issues could be mitigated by rolling the option position more frequently rather than only once per month. More frequent rolling would keep the traded contracts closer to at the money and could make the volatility signal, gamma exposure, and time-to-maturity profile more consistent over time.

7 Conclusion

This report has examined whether the volatility risk premium embedded in equity options can be systematically harvested through delta-hedged ATM straddles. Four signal specifications were designed and backtested across major U.S. ETFs and individual equities: a fixed-threshold rule (`Agent_hardThreshold`), a percentile-based rule (`Agent_Perc`), a GJR-GARCH forecast (`Agent_Garch`), and a long-horizon realized-volatility benchmark (`Agent_LongTerm`).

7.1 Key Empirical Findings

In a frictionless setting, all four agents produce positive risk-adjusted returns on SPY, with Sharpe ratios above 1.0 and maximum drawdowns contained below 15%. The fixed-threshold and percentile agents deliver the best overall risk-adjusted profile, while the long-horizon agent achieves competitive returns at the cost of higher realized volatility. PnL attribution confirms that vega is the dominant return driver across all strategies, validating that the portfolio is capturing volatility mispricing rather than unintended directional exposure. The delta-hedging overlay successfully suppresses first-order market risk, though residual contributions from higher-order Greeks such as Vanna and Volga highlight the practical limits of discrete rebalancing.

Results on individual equities are more mixed. Higher idiosyncratic jump risk and wider bid-ask spreads compress the signal-to-noise ratio, and performance varies considerably across tickers. ETFs consistently outperform single stocks, in line with their superior liquidity and lower tail risk.

7.2 The Constraint of Execution Costs

The most important practical finding is the strategy's extreme sensitivity to execution frictions. A modest 0.1% slippage assumption is sufficient to turn every agent's SPY performance from profitable to loss-making, with Sharpe ratios turning sharply negative. The core reason is mechanical: frequent delta rebalancing accumulates transaction costs quickly, and those costs erode the volatility premium before it can be realized. This result implies that frictionless backtest numbers should be treated strictly as signal-quality benchmarks rather than as achievable returns.

An unhedged straddle experiment further illustrates the trade-off. Removing the hedge dramatically raises returns but introduces directional risk and drawdowns of up to 91%, making the unhedged approach unsuitable for most institutional mandates.

7.3 Practical Relevance and Future Directions

Even if direct trading of the strategy is challenging due to execution costs, the underlying signal framework retains practical value. It provides a disciplined, quantitative way to evaluate whether current option prices are rich or cheap relative to realized volatility benchmarks. Practitioners can apply the same signal as a pre-trade sanity check—for example, before buying options ahead of an earnings event—to assess whether implied volatility is already excessively elevated. Another implementation path is to express the view through volatility-linked instruments such as VIX derivatives, where the volatility risk premium may be accessed with a different cost and hedging profile.

Future work could focus on several directions: optimizing the roll schedule so that the strategy reopens a fresh at-the-money option position at each rebalance date, and reducing hedge frequency to lower turnover costs. Since delta hedging already serves as a core risk-control mechanism by suppressing first-order directional exposure, the strategy is not purely speculative; however, additional overlays such as stop-loss rules could be introduced to further control tail losses during stress episodes. Another natural extension is to combine the current signals into a single decision framework—for example, through weighted averaging, voting rules, or a meta-model—so that information from realized-volatility, percentile-based, and GARCH-style forecasts can be processed jointly rather than traded as separate standalone rules. The same combined framework could then be tested in other asset classes where the volatility risk premium may be more accessible after costs.

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A Full Single-Stock Tables

Table 9 and Table 10 provide the complete single-stock results used to summarize this subsection.

Table 9: Full single-stock performance statistics by ticker and agent.

Ticker	Agent	Win Rate	Sharpe	Sortino	Ann. Return	Ann. Vol.	Max DD
AAPL	Agent_hardThreshold	73.17%	0.3969	0.2537	8.21%	19.88%	-21.25%
AAPL	Agent_Perc	72.22%	0.1404	0.0941	2.95%	20.74%	-26.44%
AAPL	Agent_Garch	73.33%	0.1402	0.0891	2.87%	20.16%	-23.40%
AAPL	Agent_LongTerm	86.67%	0.5435	0.3926	9.08%	15.99%	-16.86%
ACMR	Agent_hardThreshold	55.56%	-0.0069	-0.0065	-0.27%	39.16%	-25.42%
ACMR	Agent_Perc	61.54%	0.5744	0.5040	58.52%	80.20%	-32.10%
ACMR	Agent_Garch	60.00%	0.4892	0.4964	24.98%	45.58%	-31.94%
ACMR	Agent_LongTerm	60.00%	-0.9640	-0.4432	-48.66%	69.16%	-42.17%
AMD	Agent_hardThreshold	71.43%	-0.4379	-0.2366	-26.84%	71.39%	-65.30%
AMD	Agent_Perc	64.71%	-0.4454	-0.2423	-27.12%	71.04%	-66.78%
AMD	Agent_Garch	71.11%	-0.4153	-0.2214	-25.64%	71.35%	-63.65%
AMD	Agent_LongTerm	85.71%	-0.0922	-0.0424	-6.12%	68.56%	-59.33%
GOOG	Agent_hardThreshold	79.49%	1.6039	1.9659	58.84%	28.85%	-11.57%
GOOG	Agent_Perc	81.58%	1.1861	1.3496	50.25%	34.32%	-16.75%
GOOG	Agent_Garch	72.09%	1.5513	2.0530	62.73%	31.39%	-11.57%
GOOG	Agent_LongTerm	76.32%	0.9161	1.0031	32.45%	30.68%	-22.61%
INTC	Agent_hardThreshold	77.50%	1.4840	2.3184	124.20%	54.40%	-29.63%
INTC	Agent_Perc	71.05%	1.3589	2.8220	108.55%	54.09%	-24.79%
INTC	Agent_Garch	72.00%	1.4840	2.3462	129.95%	56.11%	-27.41%
INTC	Agent_LongTerm	76.09%	1.0758	0.9528	47.21%	35.94%	-28.73%
META	Agent_hardThreshold	65.79%	-0.1857	-0.1634	-5.96%	33.10%	-32.14%
META	Agent_Perc	65.00%	-0.3303	-0.2649	-9.86%	31.41%	-36.48%
META	Agent_Garch	65.79%	0.1389	0.1267	4.79%	33.72%	-28.72%
META	Agent_LongTerm	58.33%	-0.0927	-0.0777	-2.67%	29.20%	-24.03%
TSLA	Agent_hardThreshold	68.89%	0.7805	0.7674	29.31%	32.93%	-25.02%
TSLA	Agent_Perc	63.64%	0.5477	0.5530	18.19%	30.51%	-27.21%
TSLA	Agent_Garch	71.43%	0.7275	0.7003	23.56%	29.08%	-23.68%
TSLA	Agent_LongTerm	72.34%	0.1001	0.0701	2.23%	21.99%	-20.69%

Table 10: Full single-stock Greeks attribution percentages by ticker and agent.

Ticker	Agent	Delta	Gamma	Vega	Theta	Vanna	Volga	Rho	Residual
AAPL	Agent_hardThreshold	+17.52%	-30.20%	+26.97%	+10.41%	+3.20%	-1.70%	-0.01%	-9.99%
AAPL	Agent_Perc	+9.52%	-11.16%	+21.49%	+5.86%	+4.14%	+15.59%	-0.01%	-32.22%
AAPL	Agent_Garch	+9.79%	-21.20%	+28.28%	+8.64%	+3.17%	+7.35%	-0.00%	-21.56%
AAPL	Agent_LongTerm	+10.77%	-8.78%	+46.82%	-1.95%	+1.41%	-1.72%	+0.00%	-28.56%
ACMR	Agent_hardThreshold	+6.44%	-7.34%	+55.56%	-8.52%	+1.03%	+1.92%	+0.00%	-19.19%
ACMR	Agent_Perc	+23.47%	-6.99%	+48.71%	-6.22%	-6.62%	-4.91%	+0.02%	-3.07%
ACMR	Agent_Garch	-7.58%	+6.03%	+67.53%	-6.54%	+1.89%	+1.11%	+0.02%	-9.31%
ACMR	Agent_LongTerm	-1.19%	+36.75%	+5.91%	-15.02%	+10.63%	-9.47%	+0.06%	-20.97%
AMD	Agent_hardThreshold	+7.15%	-21.53%	+8.63%	+8.58%	+3.72%	+25.47%	+0.00%	-24.91%
AMD	Agent_Perc	+6.02%	-13.35%	+8.19%	+4.76%	+5.05%	+30.28%	-0.00%	-32.35%
AMD	Agent_Garch	+8.88%	-22.23%	+13.71%	+7.24%	+4.27%	+20.53%	+0.00%	-23.15%
AMD	Agent_LongTerm	-5.72%	-13.49%	+50.14%	+5.77%	-1.82%	+7.03%	+0.00%	-16.03%
GOOG	Agent_hardThreshold	+12.85%	-18.27%	+21.75%	+13.61%	-1.79%	+11.25%	-0.01%	-20.46%
GOOG	Agent_Perc	+11.50%	-2.96%	+24.81%	+2.14%	-1.25%	+20.70%	-0.01%	-36.61%
GOOG	Agent_Garch	+14.37%	-15.67%	+19.82%	+9.32%	+0.16%	+14.76%	-0.01%	-25.89%
GOOG	Agent_LongTerm	+17.89%	-2.98%	+26.98%	+3.76%	+0.17%	+11.49%	-0.01%	-36.71%
INTC	Agent_hardThreshold	-5.17%	+7.87%	+20.40%	+10.16%	+2.78%	+20.65%	-0.00%	-32.97%
INTC	Agent_Perc	+5.30%	+11.22%	+14.44%	-1.52%	+3.31%	+24.23%	-0.00%	-39.98%
INTC	Agent_Garch	+0.85%	-0.51%	+31.48%	+8.82%	+4.69%	+16.82%	-0.00%	-36.82%
INTC	Agent_LongTerm	-10.42%	+10.34%	+42.03%	+3.15%	+5.71%	+3.51%	-0.00%	-24.83%
META	Agent_hardThreshold	-19.06%	-24.77%	+9.40%	+27.99%	+0.25%	-3.99%	+0.02%	+14.53%
META	Agent_Perc	-26.31%	-20.21%	+26.12%	+9.76%	+0.01%	+16.33%	+0.02%	-1.24%
META	Agent_Garch	-17.56%	-27.06%	+9.44%	+33.54%	-0.02%	+0.73%	+0.01%	+11.65%
META	Agent_LongTerm	+6.50%	-48.47%	+10.06%	+22.45%	+2.51%	+2.28%	+0.00%	+7.72%
TSLA	Agent_hardThreshold	+5.24%	-13.41%	+50.83%	+2.63%	+2.89%	-24.38%	+0.00%	+0.61%
TSLA	Agent_Perc	+1.06%	+10.34%	+32.66%	-10.97%	+2.58%	+8.57%	+0.00%	-33.79%
TSLA	Agent_Garch	-2.72%	+1.94%	+53.48%	-4.59%	+3.01%	+1.40%	+0.00%	-32.85%
TSLA	Agent_LongTerm	-11.85%	+5.65%	+41.71%	-3.36%	+3.69%	+5.16%	-0.00%	-28.58%

B SPY Execution-Frictions Diagnostics

Figure 6 presents the SPY slippage diagnostics used to assess how execution frictions affect the strategy.

Figure 6: SPY slippage diagnostics.

C SPY Unhedged Diagnostics

Figure 7 presents the SPY unhedged-straddle diagnostics used to compare the no-hedge variant with the delta-hedged strategy.

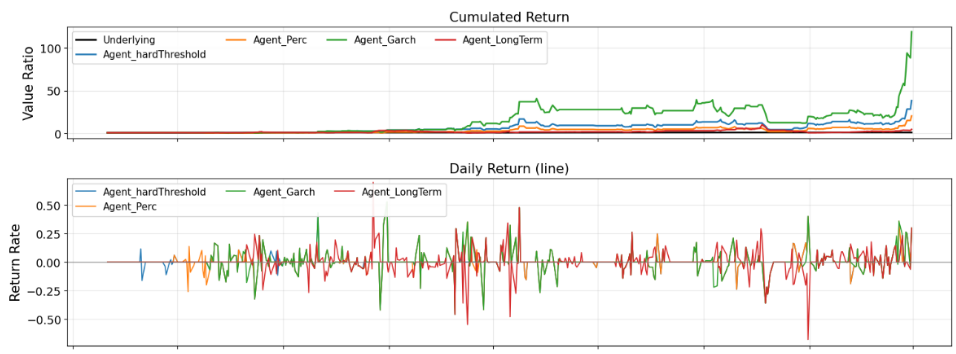

Figure 7: SPY unhedged-straddle diagnostics.

D Greeks Attribution: Dollar PnL Time Series

Figure 8 shows the daily dollar PnL contribution of each Greek component across the full backtest period. Each panel corresponds to one strategy agent and decomposes daily total PnL into delta, gamma, vega, theta, vanna, volga, rho, and residual buckets. This time-series view complements the pie chart by showing when each risk factor dominated and when attribution spikes occurred.

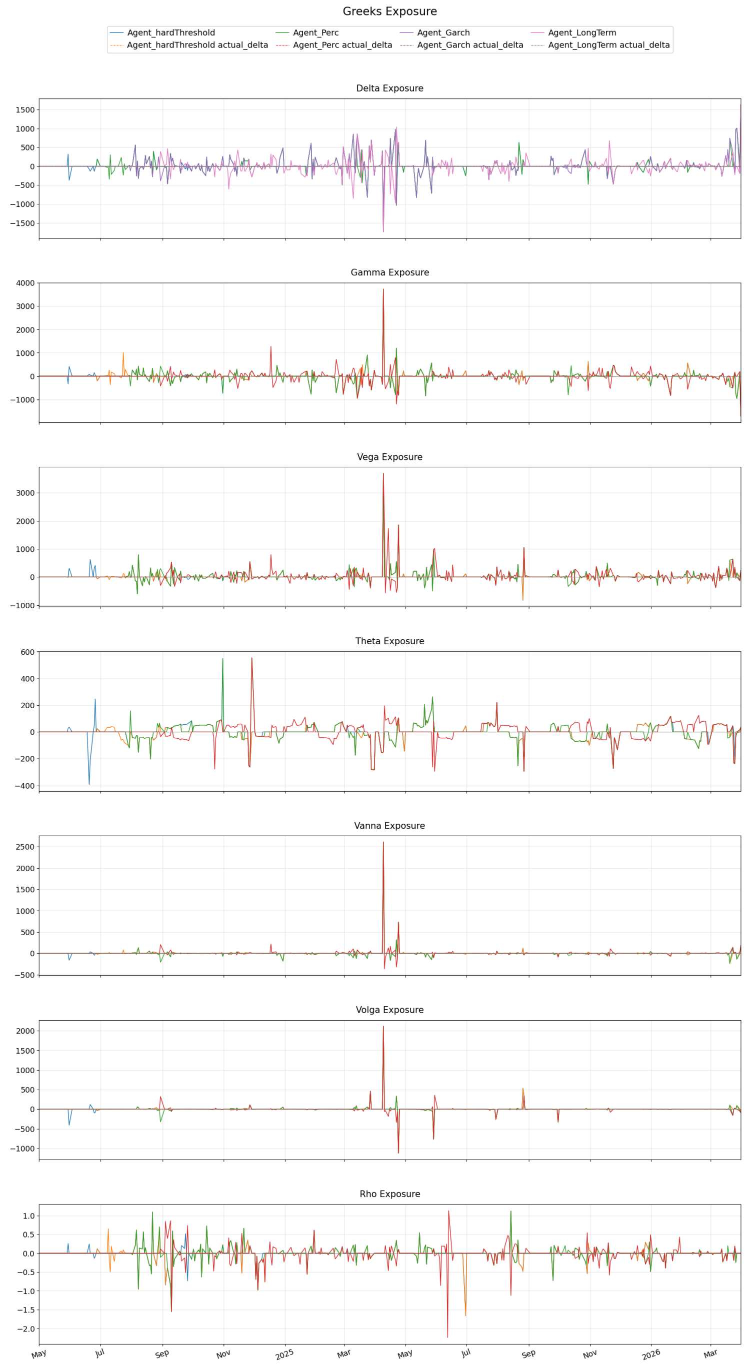

Figure 8: Daily dollar PnL attribution by Greek component for each agent on SPY.

E Greeks Exposure Over Time

Figure 9 shows the evolution of Greeks exposure over the backtest period. Unlike the dollar-attribution series, this plot tracks the raw sensitivity levels (e.g. net vega, net delta, and net gamma) as the position changes, providing insight into how risk exposures build and unwind across entry, holding, and exit phases.

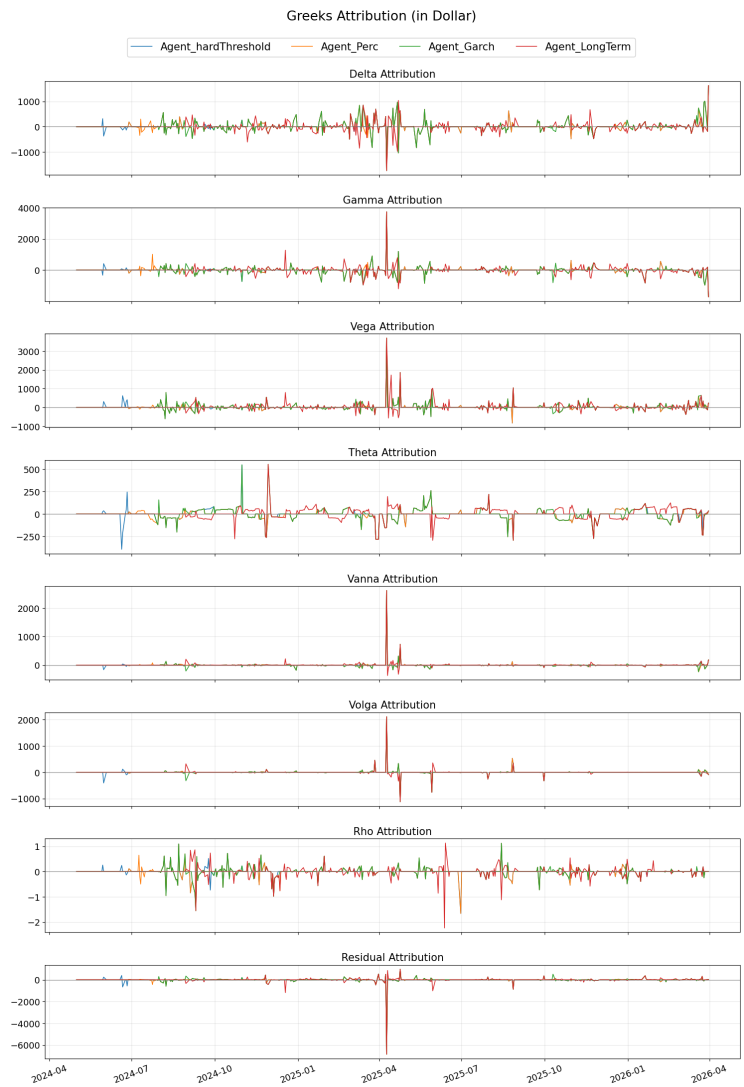

Figure 9: Greeks exposure over time for each agent on SPY.

F QQQ: Return and Greeks

Figure 10 and Figure 11 report the return and Greek diagnostics for QQQ.

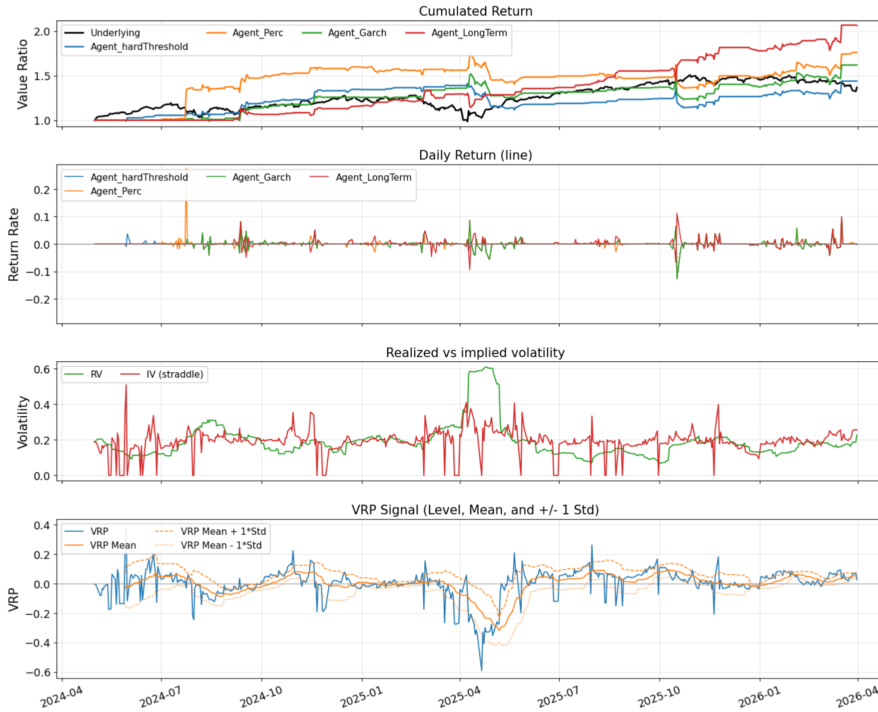

Figure 10: QQQ return diagnostics.

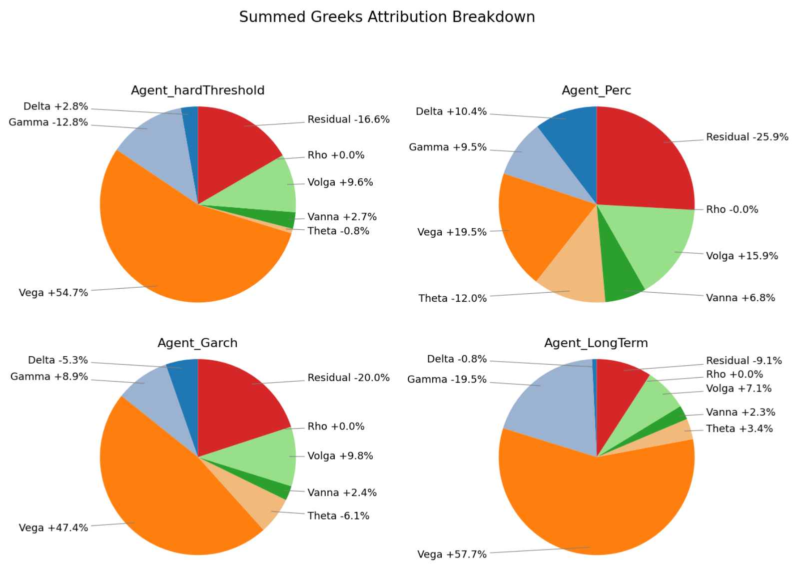

Figure 11: QQQ Greeks diagnostics.

G DIA: Return and Greeks

Figure 12 and Figure 13 report the return and Greek diagnostics for DIA.

Figure 12: DIA return diagnostics.

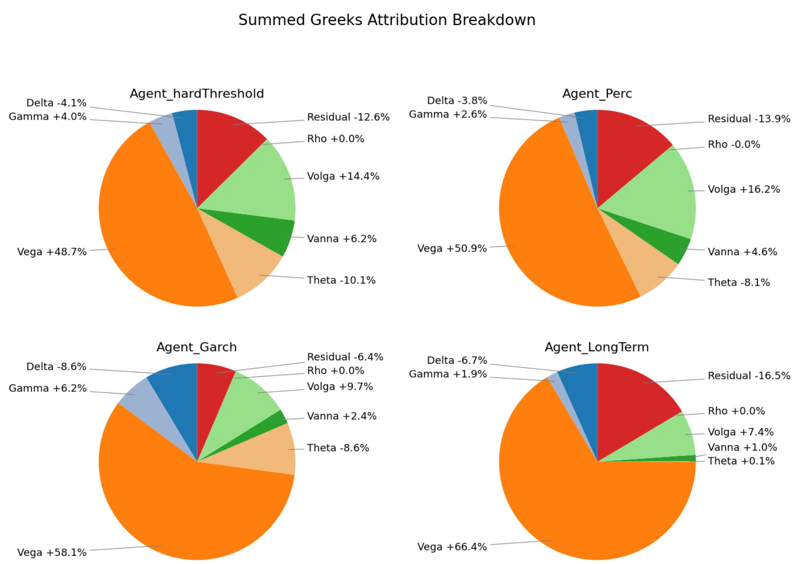

Figure 13: DIA Greeks diagnostics.